

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

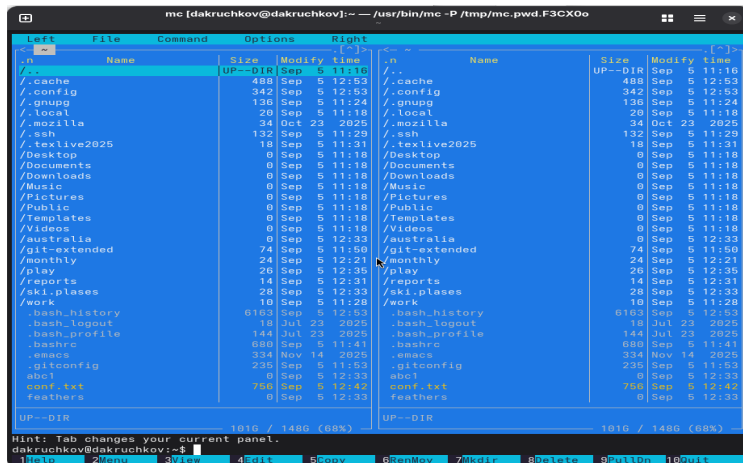


Рисунок 1: Запуск mc

Работа с Midnight Commander

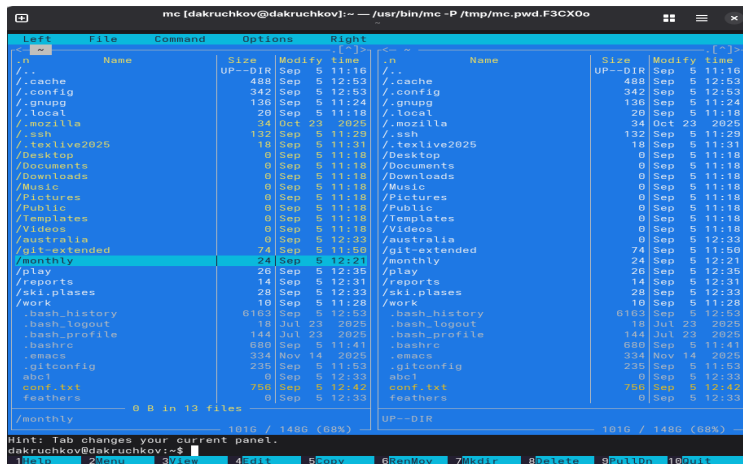


Рисунок 2: Выделение

Работа с Midnight Commander

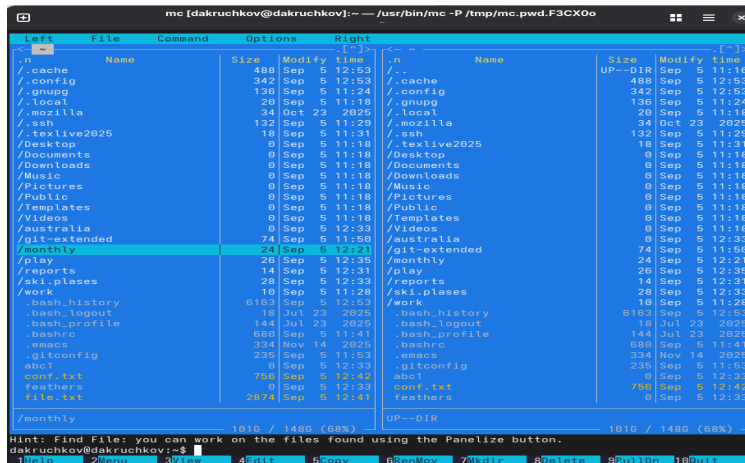


Рисунок 3: Отмена

Работа с Midnight Commander

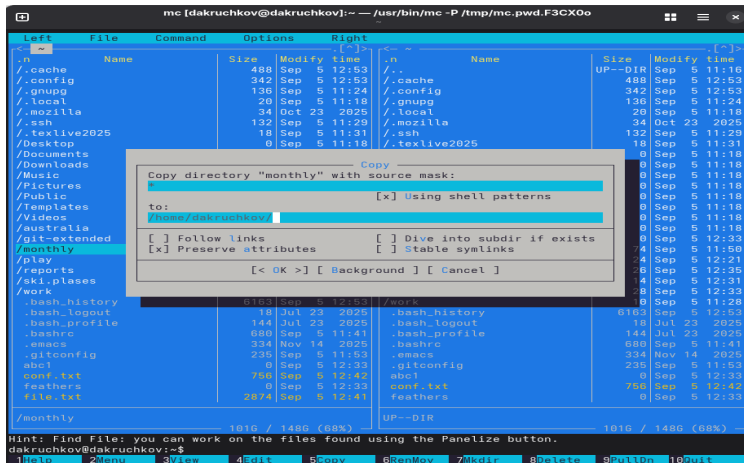


Рисунок 4: Копирование

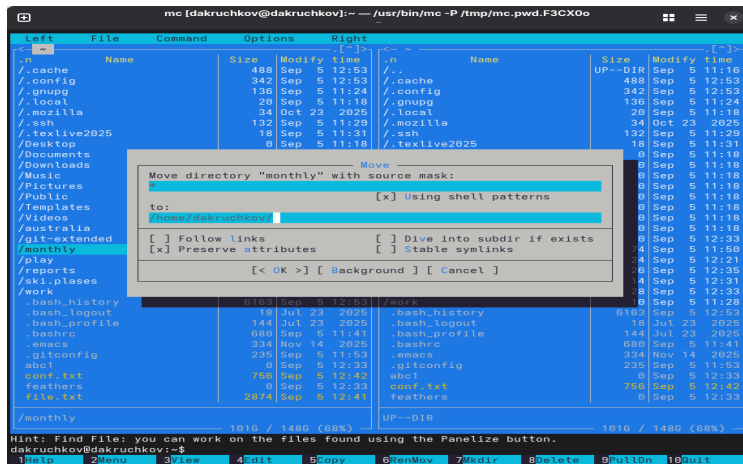


Рисунок 5: Перемещение

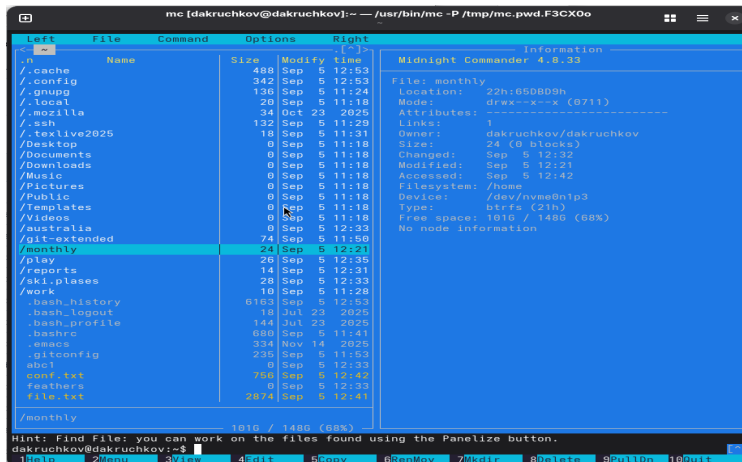


Рисунок 6: Информация

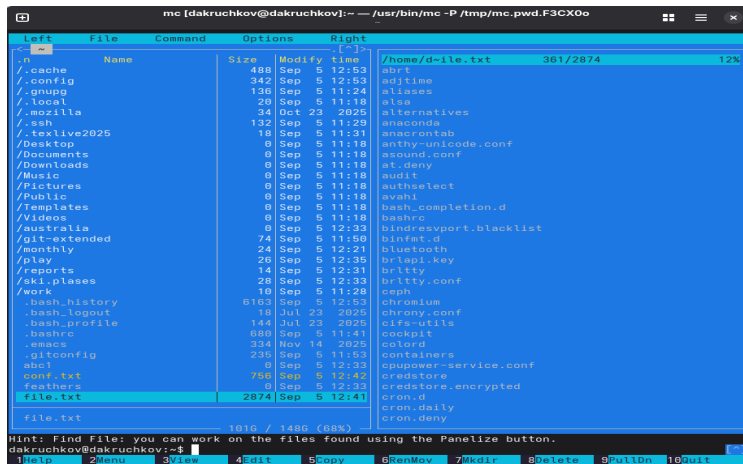


Рисунок 7: Быстрый просмотр

The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'dakruchkov' on a system named 'dakruchkov', with the command line showing the path '/usr/bin/mc -P /tmp/mc.pwd.F3CX0o'. The interface is divided into several panels:

- Left Panel:** A list of files and directories. The file 'file.txt' is highlighted in blue.
- File Panel:** Displays the details of the selected file 'file.txt'.
- Command Panel:** Shows the command 'file.txt' and the disk usage '101G / 148G (68%)'.
- Options Panel:** Displays the command 'file.txt'.
- Right Panel:** Displays detailed information about the selected file.

Left	File	Command	Options	Right
File list:	file.txt	file.txt	file.txt	Information

File Information:

- File: file.txt
- Location: 22h:650D6Fh
- Mode: -rw-r--r-- (0644)
- Attributes: -----
- Links: 1
- Owner: dakruchkov/dakruchkov
- Size: 2874 (8 blocks)
- Changed: Sep 5 12:41
- Modified: Sep 5 12:41
- Accessed: Sep 5 12:41
- Filesystem: /home
- Device: /dev/nvme0n1p3
- Type: btrfs (21h)
- Free space: 101G / 148G (68%)
- No node information

File List (Left Panel):

Name	Size	Modify	time
..			
./			
./cache	488	Sep 5 12:53	
./config	342	Sep 5 12:53	
./gnupg	136	Sep 5 11:24	
./local	20	Sep 5 11:18	
./mozilla	34	Oct 23 2025	
./ssh	132	Sep 5 11:29	
./texlive2025	18	Sep 5 11:31	
/Desktop	0	Sep 5 11:18	
/Documents	0	Sep 5 11:18	
/Downloads	0	Sep 5 11:18	
/Music	0	Sep 5 11:18	
/Pictures	0	Sep 5 11:18	
/Public	0	Sep 5 11:18	
/Templates	0	Sep 5 11:18	
/Videos	0	Sep 5 11:18	
/australia	0	Sep 5 12:33	
/git-extended	74	Sep 5 11:50	
/monthly	24	Sep 5 12:21	
/play	26	Sep 5 12:35	
/reports	14	Sep 5 12:31	
/ski.plases	28	Sep 5 12:33	
/work	10	Sep 5 11:28	
./bash_history	6163	Sep 5 12:53	
./bash_logout	18	Jul 23 2025	
./bash_profile	144	Jul 23 2025	
./bashrc	680	Sep 5 11:41	
./emacs	334	Nov 14 2025	
./gitconfig	235	Sep 5 11:53	
abcl	0	Sep 5 12:33	
conf.txt	766	Sep 5 12:42	
feathers	0	Sep 5 12:33	
file.txt	2874	Sep 5 12:41	

Hint: Find File: you can work on the files found using the Panelize button.

dakruchkov@dakruchkov:~\$

1 Help 2 Menu 3 View 4 Edit 5 Copy 6 RenMov 7 Mkdir 8 Delete 9 PullDn 10 Quit

Рисунок 8: Информация

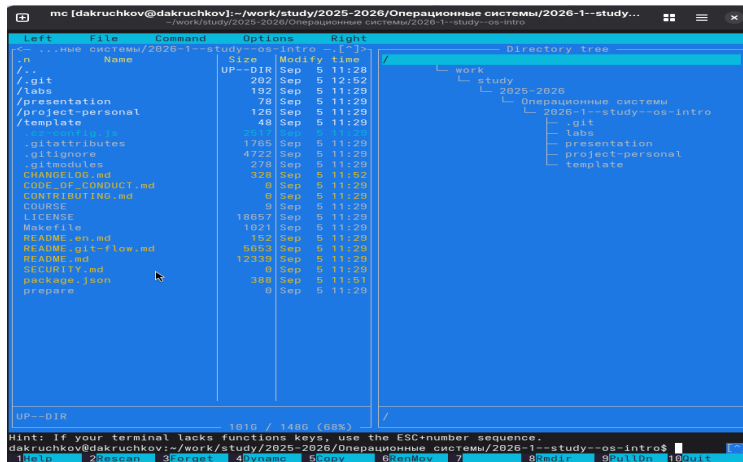


Рисунок 9: Дерево каталогов



```
mc [dakruchkov@dakruchkov]:~/work/study/2025-2026/Операционные системы/2026-1--study...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro

/home/dakruchkov/work/study/2025-~/2026-1--study--os-intro/Makefile 1021/1021 100%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
    @echo 'Usage:'
    @echo '  make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
| awk 'BEGIN {FS = ".*?###"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n"
, $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {F
S = ".*?###"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @echo

list:    ## List of courses
    @. ./template/config/script/list-courses

prepare:    ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule:    ## Update submodules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); gi
t reset --hard origin/$$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; git clea
n -dfx'
```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander

```
mc [dakruchkov@dakruchkov]:~/work/study/2025-2026/Операционные системы/2026-1--study...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
Makefile [----] 0 L: [ 1+ 0 1/ 28] *(0 /1021b) 0083 0x053 [+][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo ' make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n"
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {F

list:<-->## List of courses
----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<---->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$$(git rev-parse --abbrev-ref HEAD); gi

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

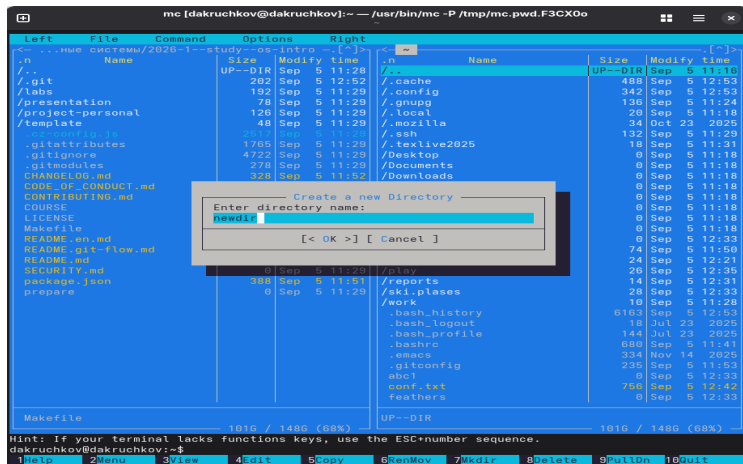


Рисунок 12: Создание каталога

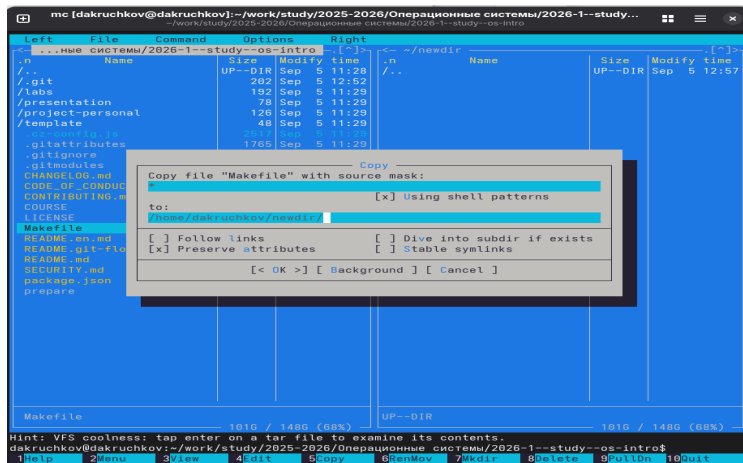


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

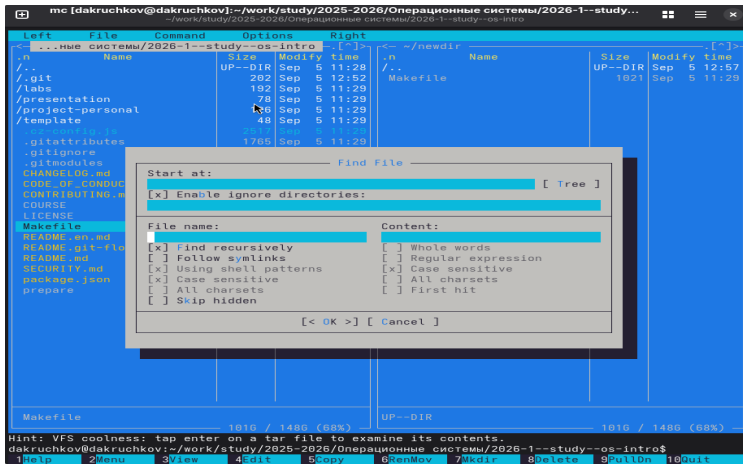


Рисунок 14: Поиск файлов

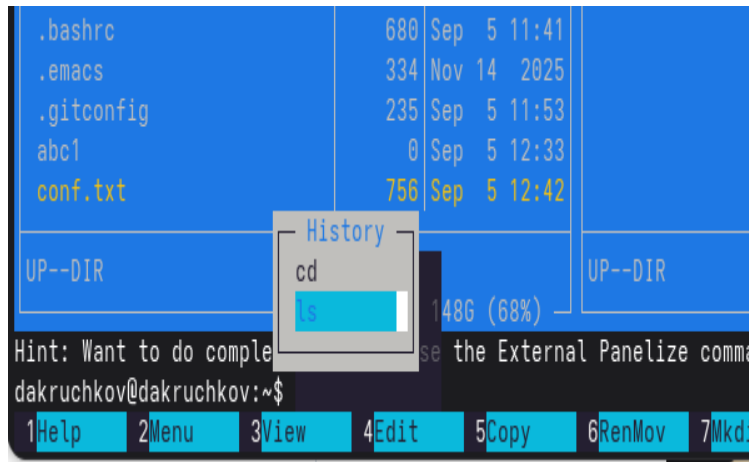


Рисунок 15: История команд

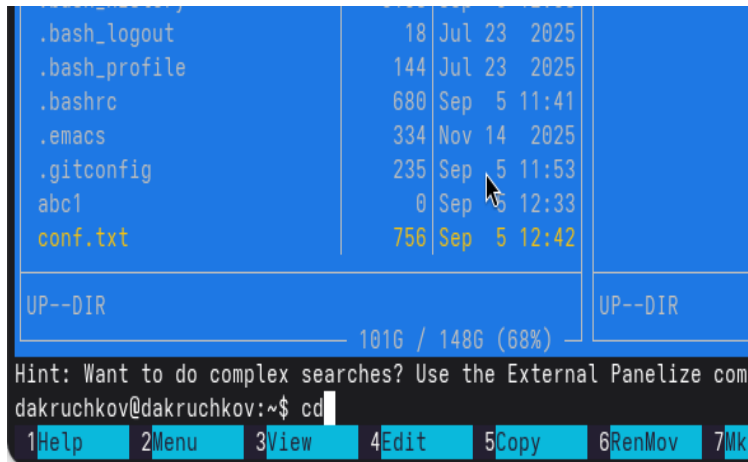


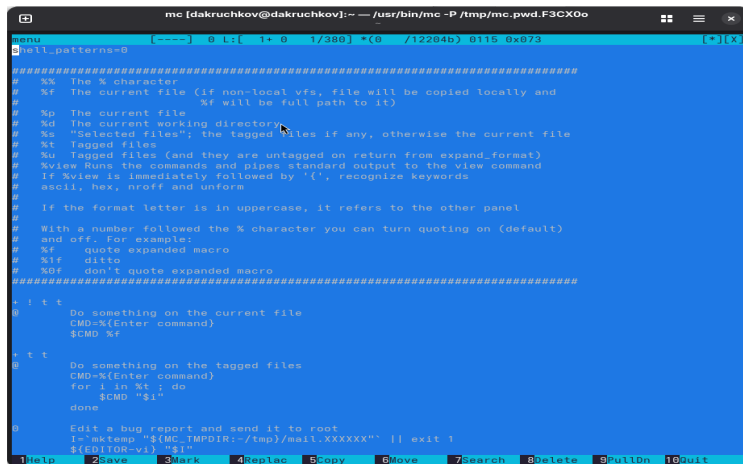
Рисунок 16: Переход в домашний каталог



The screenshot shows a terminal window titled "mc [dakruchkov@dakruchkov]:~ — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o". The window displays the contents of the "mc.ext.ini" file, which is the Midnight Commander 4.8 extension file. The text is as follows:

```
mc.ext.ini [----] 0 L: [ 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F8-c-e). If you edit this file in any other way
# while mc is running, you will need to press F8-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxxx] can be referenced as "Include=xxxxx" from other sections.
# Section [Include/xxxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
#
# A section describing a file can contain following keys:
#
#   File descriptions:
#
#       Directory
#           Matches any directory matching regular expression.
#
# 1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullOn 10Quit
```

Рисунок 17: Просмотр файла расширений

The image shows a terminal window with the Midnight Commander (MC) interface. The title bar indicates the user is 'dakruchkov' and the current directory is '/usr/bin/mc'. The main window displays the 'menu' screen, which lists various file operations and their shortcuts. The menu is organized into sections: a top section with file character and path macros, a middle section with file and directory macros, and a bottom section with file and directory macros. The bottom of the screen features a status bar with numbered shortcuts: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, and 10Quit. The background is blue, and the text is white.

```
mc [dakruchkov@dakruchkov:~ — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o
menu [----] 0 L: 1+ 0 1/380 * (0 /12204b) 0115 0x073 [*][X]
shell_patterns=0
#####
# % The % character
# %f The current file (if non-local vfs, file will be copied locally and
# %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '(', recognize keywords
# ascll, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %lf ditto
# %0f don't quote expanded macro
#####
+ ! t t
0 Do something on the current file
  CMD=%{Enter command}
  $CMD %f
+ t t
0 Do something on the tagged files
  CMD=%{Enter command}
  for i in %t ; do
    $CMD "$i"
  done
0 Edit a bug report and send it to root
  I=mktemp "${MC_TMPDIR:-/tmp}/mail.XXXXXX" || exit 1
  $(EDITOR-vi) "$i"
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 18: Просмотр файла меню

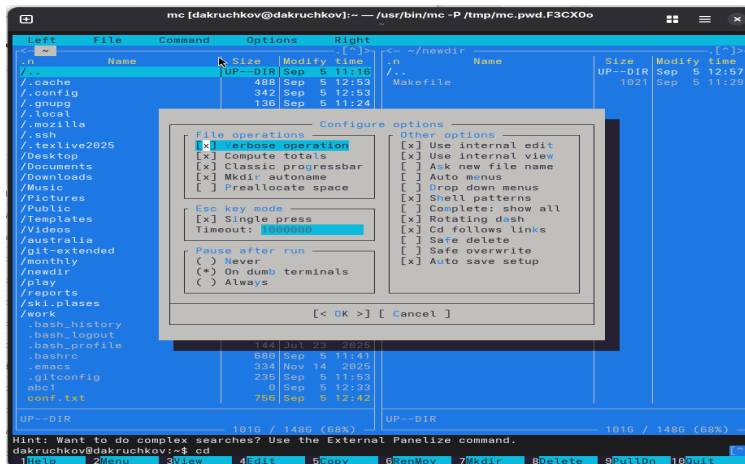


Рисунок 19: Конфигурация

Работа с Midnight Commander

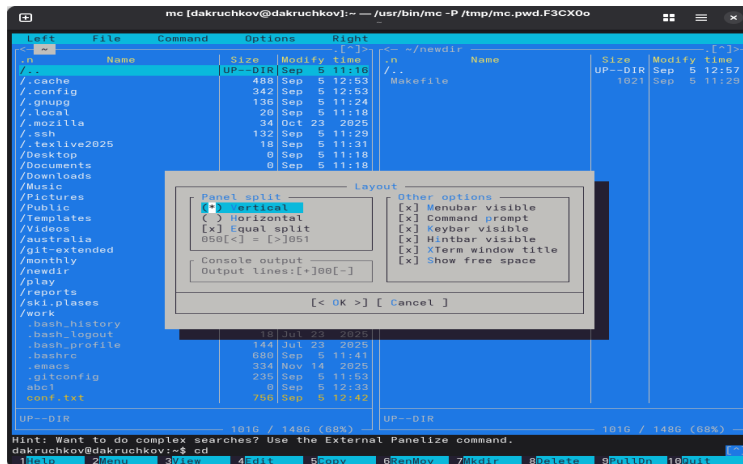


Рисунок 20: Внешний вид

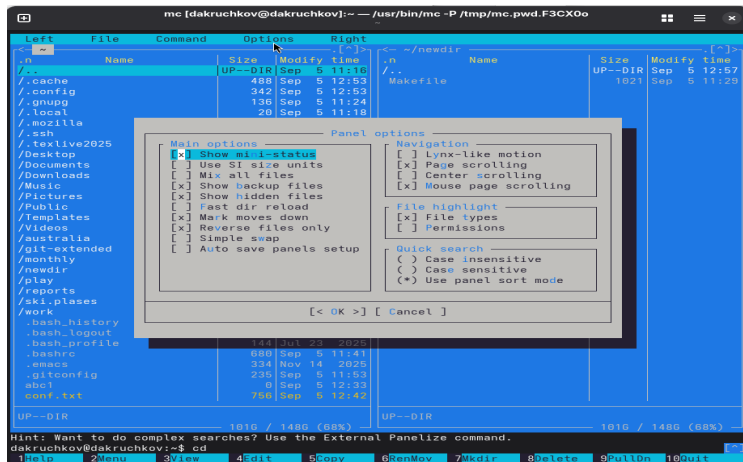


Рисунок 21: Настройки панелей

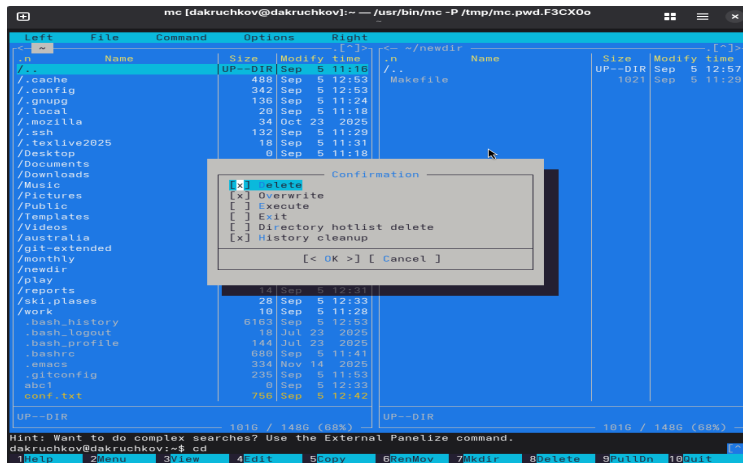


Рисунок 22: Подтверждение

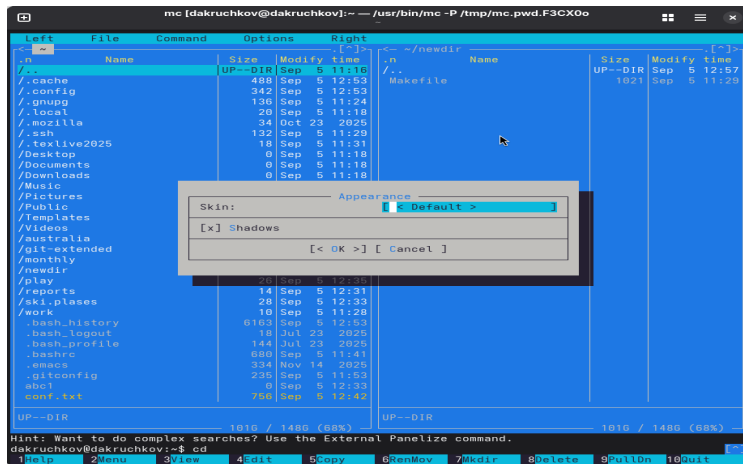


Рисунок 23: Оформление

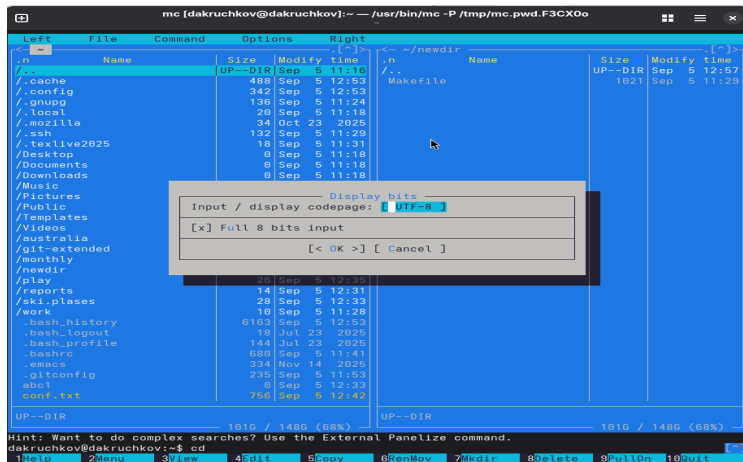


Рисунок 24: Кодировка символов

Работа с Midnight Commander

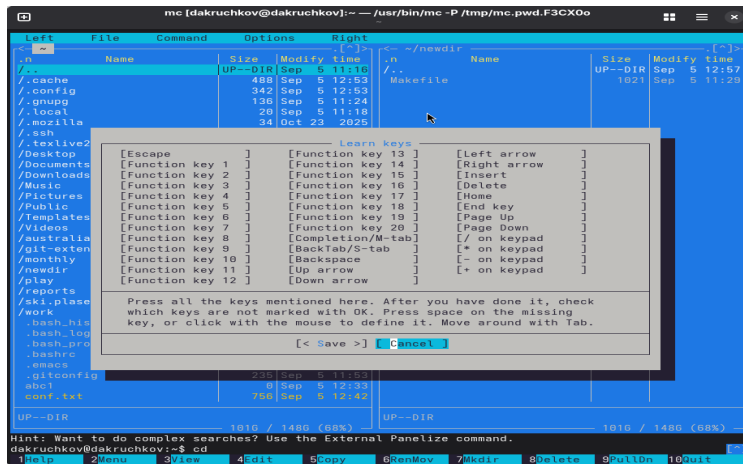


Рисунок 25: Распознавание клавиш

Работа с редактором Midnight Commander

```
mc [dakruchkov@dakruchkov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o
~ /newdir
Makefile [-----] 34 L: [ 1+24 25/ 28] *(776 /1021b) 0010 0x00A [~][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1=
<----->@grep -E '###.*' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n"
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {F
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

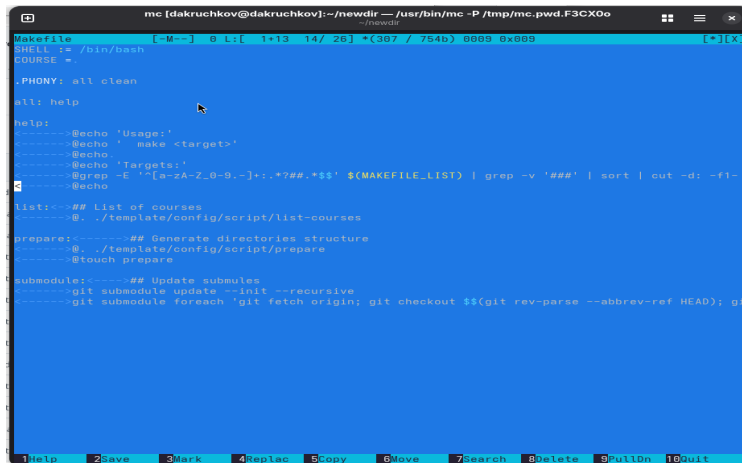
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); g

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'dakruchkov' at 'dakruchkov', and the current directory is '~/newdir'. The file being edited is 'Makefile'. The editor shows the following content:

```
Makefile [-M--] 0 L: [ 1+13 14/ 26] *(307 / 754b) 0000 0x000 [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9.-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

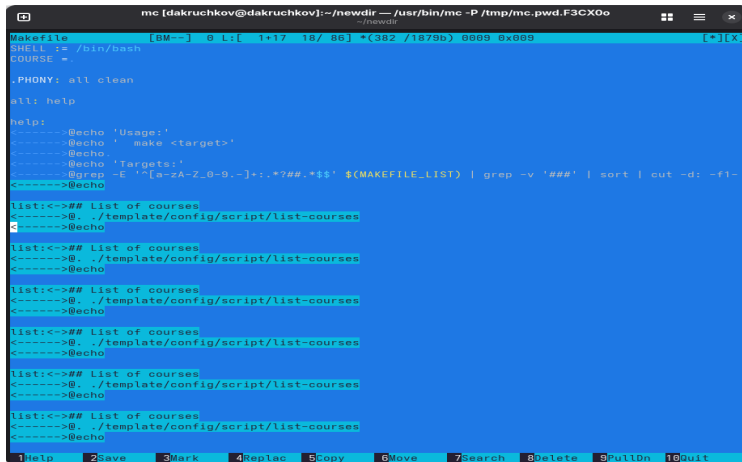
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); gi
```

At the bottom of the window, there is a menu bar with the following options: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander



```
mc [dakruchkov@dakruchkov]: ~/newdir — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o
~/newdir
Makefile [BM--] 0 L: 1+17 18/ 86] *(382 /1879b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo ' make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->### List of courses
----->@. ./template/config/script/list-courses
----->@echo

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

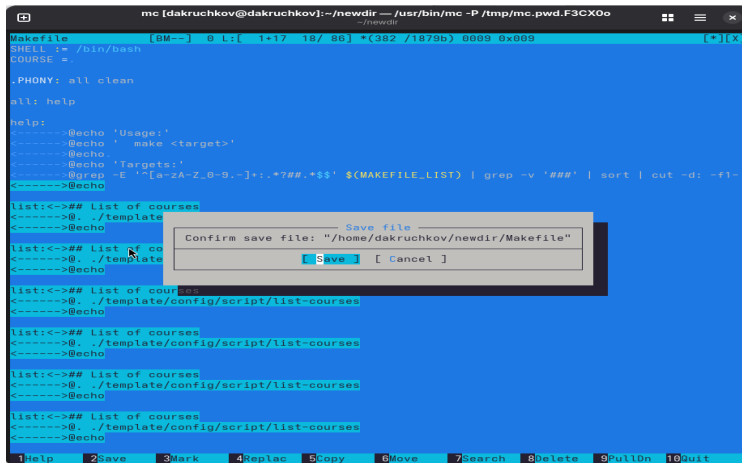


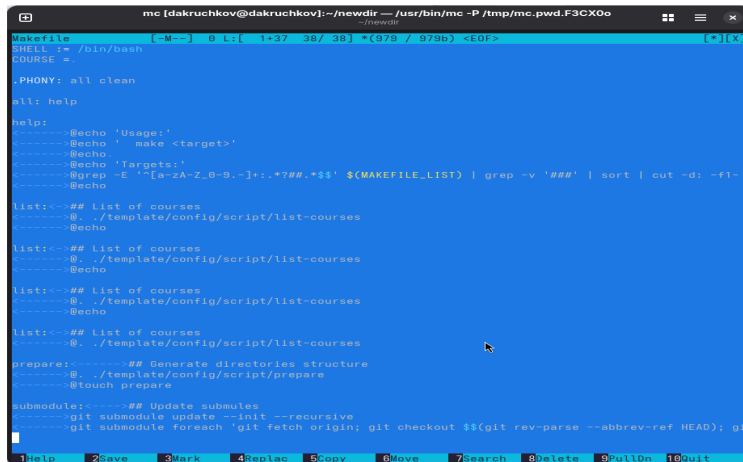
Рисунок 29: Сохранение

Работа с редактором Midnight Commander

```
mc [dakruchkov@dakruchkov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o
~/newdir
Makefile [BM--] 0 L: [ 1+17 18/ 38] *(382 / 979b) 0009 0x009 [*][X]
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
----->@echo 'Usage:'
----->@echo '  make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.~+.:?#$.*$]' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
----->@echo
list:<-->## List of courses
----->@. ./template/config/script/list-courses
----->@echo
list:<-->## List of courses
----->@. ./template/config/script/list-courses
----->@echo
list:<-->## List of courses
----->@. ./template/config/script/list-courses
----->@echo
list:<-->## List of courses
----->@. ./template/config/script/list-courses
----->@echo
prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare
submodule:<---->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

Рисунок 30: Отмена

Работа с редактором Midnight Commander



```
mc [dakruchkov@dakruchkov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.F3CX0o
~/newdir
Makefile  [-M--]  0 L:[ 1+37 38/ 38] *(978 / 979b) <EOF>  [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean
all: help

help:
----->@echo 'Usage:'
----->@echo ' make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

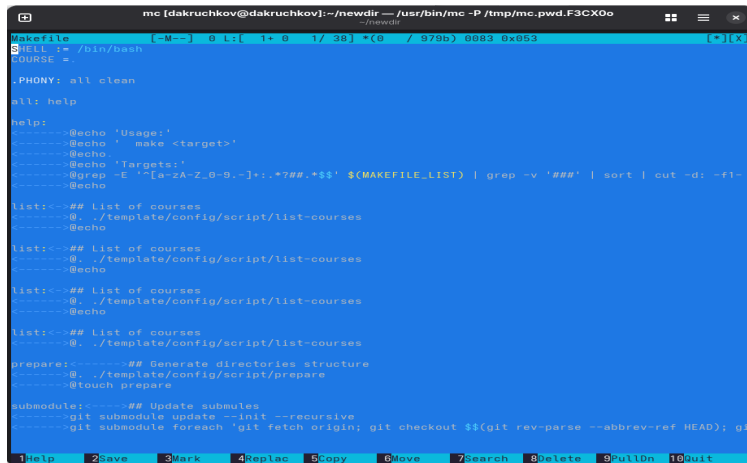
list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<---->## Update submules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); gi
```

Рисунок 31: Переход в конец файла

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is 'dakruchkov' and the current directory is '/usr/bin/mc'. The main window displays a file named 'Makefile' with the following content:

```
Makefile [-M--] 0 L: [ 1+ 0 1/ 38] *(0 / 979b) 0083 0x053 [*][X]
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
----->@echo 'Usage:'
----->@echo ' make <target>'
----->@echo
----->@echo 'Targets:'
----->@grep -E '^[a-zA-Z_0-9.-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1=
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

list:<->## List of courses
----->@. ./template/config/script/list-courses
----->@echo

prepare:<----->## Generate directories structure
----->@. ./template/config/script/prepare
----->@touch prepare

submodule:<----->## Update submodules
----->git submodule update --init --recursive
----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git
```

At the bottom of the window, there is a command palette with the following options: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander

[illegible]

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

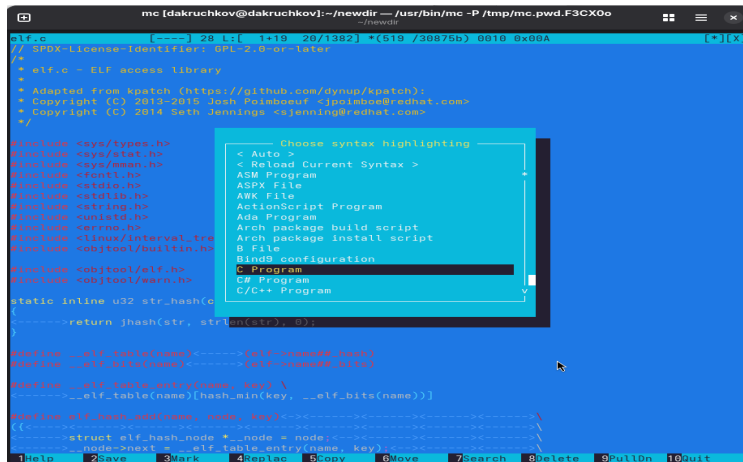


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.